WEEKLY LESSON PLAN · 2026-27

Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School
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SEMESTER 1
Week 1

MONDAY — Day 1 - Procedures, Lab Safety & Design Notebook

Aug 10

OBJECTIVE
Students will classify manufacturing-lab hazards, apply the NIOSH hierarchy of controls to select the most effective control for a given task, complete a Job Safety Analysis (JSA) with a likelihood x severity risk score, and set up a dated, numbered engineering design notebook as a professional record.

TEKS
§127.813(c)(1) safety, professional standards & responsible tool use

ELPS / EB SUPPORT
ELPS/EB supports at grade level: hazard-class icons and a live tool demo (VC, DP); pre-teach hierarchy-of-controls terms with a labeled pyramid before use (PV); JSA drafted with a bilingual peer and sentence stems for the 'control + justification' cell (PN, CD); wait time and rephrase/slow on the risk-scoring steps (WT, RS); extra time to finish the notebook set-up (ET).

AGENDA · 45 MIN
- 5′ Bell-work: list every hazard you can spot in a shop photo
- 10′ Why safety is a professional standard + the six hazard classes
- 12′ Hierarchy of controls, machine guarding, LOTO & PPE
- 10′ Worked Job Safety Analysis + guided you-try
- 5′ The engineering notebook as a legal/professional record — set up Year-2 sections
- 3′ Exit ticket + sign safety agreement

SLIDES / ACTIVITY
Working Safely: Hazards, Controls & the Professional Record

NOTEBOOK
Set up cover, table of contents, numbered/dated pages. First entry: one thing you want to design and build this year.

TURN IN
Job Safety Analysis (JSA) + Risk Scoring, Lab-Safety Agreement & Non-Negotiables

TUESDAY

Day 2 - Manufacturing Systems

Aug 11

OBJECTIVE

Students will model production as an input -> process -> output system with a control loop, identify the six inputs for a real product, classify job/batch/repetitive/continuous production against the volume-variety product-process matrix, and compute a value-added ratio and per-unit cost to justify a production-type choice.

TEKS

`§127.813 (draft, verify strand) manufacturing systems - inputs, processes, outputs & production types`

ELPS / EB SUPPORT

ELPS/EB supports at grade level: illustrated systems diagram and a picture word-bank for the six inputs (VC, DP, PV); the video gets an active-viewing note-catcher with sentence stems (CD); classification and the VA-ratio steps done in pairs so EB students reason aloud (PN); slow/rephrase the economies-of-scale graph and allow extra time on the calculation (RS, WT, ET).

AGENDA · 45 MIN

- **5′** Bell-work: how is making a car different from a wedding cake?
- **8′** The systems model + the six inputs + process families
- **7′** Value-added vs the 7 wastes + worked VA-ratio
- **8′** Toyota Production System video + active-viewing notes
- **7′** Production types, product-process matrix & economies of scale
- **7′** Guided practice: classify & justify + start worksheet
- **3′** Exit ticket

SLIDES / ACTIVITY

Manufacturing Systems: Inputs, Processes & Production Types

NOTEBOOK

Draw the input -> process -> output diagram. List one product for each production type (job, batch, continuous).

TURN IN

Production Systems Analysis: Inputs, Value-Added & Cost

WEDNESDAY

Day 3 - Design Process: Ask, Imagine, Plan

Aug 12

OBJECTIVE

Students will apply the engineering design process to define a real-world problem and begin designing a solution (Ask, Imagine, Plan).

TEKS

`§127.813 (draft, verify strand) production planning, logistics & workflow`

ELPS / EB SUPPORT

ELPS/EB supports at grade level: color-coded station diagram with the bottleneck highlighted (VC, DP); pre-teach cycle time / takt / throughput with a worked model before independent work (PV); worksheet calculations started with a bilingual partner and a formula reference card (PN, CD); wait time on the two-step feasibility comparison and extra time on the recompute (WT, ET, RS).

AGENDA · 45 MIN

- **5′** Bell-work: purpose of the engineering design cycle
- **16′** Present: the 5-step engineering design process
- **7′** Pick a real problem (list 3, choose 1)
- **17′** Start the design sheet: Ask -> Imagine (4 idea sketches) -> Plan (Design #1 + materials)

SLIDES / ACTIVITY

Engineering Design Process — Pick Your Problem

NOTEBOOK

Draw a workflow diagram for a simple product (e.g., a sandwich shop). Label one bottleneck.

TURN IN

Engineering Design Process sheet - Ask / Imagine / Plan

THURSDAY

Day 4 - Finish Your Design + AI Product Image

Aug 13

OBJECTIVE

Students will finish their final design, write a structured AI prompt, and generate + submit a product image.

TEKS

§127.813 (draft, verify strand) industry overview, careers & workplace safety

ELPS / EB SUPPORT

ELPS/EB supports at grade level: hazard pictograms and the NFPA color diamond taught visually with a legend (VC, DP, PV); the factory video gets an active-viewing note-catcher (CD); career research uses a template with sentence stems and a bilingual partner (CD, PN); the ROI/payback calculation modeled first, with a formula card and extra time (RS, ET); wage terms rephrased and slowed (RS, WT).

AGENDA · 45 MIN

- **8′** Bell-work + video (start 1:28): name 2 improvements and why each matters
- **10′** Model the notebook page (prompt on top, sketch on bottom) + copy the prompt
- **20′** Finish final design, then generate the product image in Gemini
- **7′** Submit the image to Google Classroom

SLIDES / ACTIVITY

Finish Your Design + Product Image

NOTEBOOK

Note one manufacturing career: what they do, pay range, and education needed.

TURN IN

AI product image (Gemini) + notebook page (prompt + final design) -> Google Classroom

FRIDAY

Day 5 - Present Your Product

Aug 14

OBJECTIVE

Students will present their product - communicating the problem it solves, their final design, and how it works - and give supportive peer feedback.

TEKS

§127.813 (draft, verify strand) engineering design process applied to a manufacturable product

ELPS / EB SUPPORT

ELPS/EB supports at grade level: the design-process loop shown as a labeled diagram and the decision matrix modeled cell-by-cell (VC, DP); pre-teach constraint / criteria / tolerance / trade-off before use (PV); problem statement and matrix built with a bilingual partner using sentence stems, and drawings accepted for concepts (PN, CD, VC); extra time and a worked template for the weighted-score arithmetic (ET, RS).

AGENDA · 45 MIN

- **5′** Bell-work: write your 4-sentence presentation script
- **8′** Model: the four things every presenter says + how to be a great audience
- **32′** Presentations: each student shows their AI product image + notebook page (~1 min each)
- **5′** Exit ticket: one thing you would design next

SLIDES / ACTIVITY

Present Your Product

NOTEBOOK

Glue in the product-idea sheet. Sketch your top idea and list what it's made of.

TURN IN

Presentation script (4 sentences) + peer-feedback sentence frames